

Dish Cost & Profit Score

Finding the high-cost and the high-margin dishes — without owners entering any costs

1. What it finds

A score for each dish that says how **expensive it is to make** and how **profitable** it is — using the fields you already have, plus two things the AI reads on its own. The owner enters no ingredient costs.

2. The factors

Measured fields (already on the dish): Price **P**, Quantity per serving **Q**, Prep time **T**. **AI-derived** from the dish name and short description — the AI knows the recipes, so it reads them for free: ingredient cost tier **G** (Low = −1, Medium = 0, High = +1) and perishability **R** (shelf-stable = 0, perishable = 1, for spoilage / wastage).

3. Put the measured fields on one scale

Standardise P, Q and T across the dishes on the menu, so each becomes a comparable number around zero (μ = the average, σ = the standard deviation):

$$z(x) = (x - \mu_x) \div \sigma_x$$

if every dish is the same ($\sigma = 0$), set $z = 0$

4. The two formulas

$$C_i = w_1 \cdot z(Q_i) + w_2 \cdot z(T_i) + w_3 \cdot G_i + w_4 \cdot R_i$$

Estimated input cost — quantity = food, prep = labour, G = ingredient tier, R = perishability

$$\Pi_i = z(P_i) - C_i$$

Profitability = standardised price – estimated cost

Suggested weights: $w_1 = 1.0$ (food / quantity), $w_2 = 0.7$ (labour / prep), $w_3 = 1.2$ (ingredient tier — the strongest cost signal), $w_4 = 0.5$ (perishability). The directions carry the meaning: a bigger portion, longer prep, or costlier / perishable ingredients all *raise* the cost C; a higher price *raises* the profit Π .

5. Reading the result

Π is **centred at zero**. Positive = a **high-margin** dish (price comfortably above what it costs); negative = a **high input-cost** / **thin-margin** dish. Rank by Π for your most profitable items; the most negative are the expensive, low-margin ones to reprice or rework. The cost score C, on its own, ranks how expensive each dish is to make.

6. Where G and R come from — with almost no typing

The owner does **not** list ingredients. The AI recognises the dish from its **name** and recalls the recipe — “Paneer Butter Masala” → paneer, butter, cream, cashew → **High** cost and **perishable**; “Idli” → rice and dal → **Low**. The short description only sharpens it. The whole menu is labelled in one pass; only a genuinely unusual dish is flagged for a single Low / Med / High tap.

7. Worked example

P, Q, T are standardised across the four dishes; G and R are read by the AI. Then $C = z(Q) + 0.7 \cdot z(T) + 1.2 \cdot G + 0.5 \cdot R$ and $\Pi = z(P) - C$.

Dish	P	Q	T	G	R	z(P)	z(Q)	z(T)	Cost C	Profit Π
Paneer Butter Masala	220	1	18	High	1	1.59	−0.58	0.52	1.48	0.11
Idli (2)	50	2	10	Low	0	−1.11	1.73	−1.15	−0.28	−0.84
Veg Biryani	120	1	22	Med	0	0.00	−0.58	1.36	0.37	−0.37
Masala Dosa	90	1	12	Med	0	−0.48	−0.58	−0.73	−1.09	0.61

Most profitable: Masala Dosa (cheap to make, fair price). **Highest input cost: Paneer Butter Masala** (premium + perishable) — and despite its ₹220 price its margin is only modest. **Idli** is the thinnest margin (low price, big portion). So the owner learns at a glance which plates earn and which merely turn over.

8. Notes

- **It is an estimate in buckets** (the ingredient tier is Low / Med / High, not exact rupees) — strongest for standard dishes; rare custom items can be confirmed with one tap.
- **It plugs into the promotion formula.** This profit score Π is the margin signal the promotion score wanted — so the AI can promote the dishes that are both wanted *and* profitable.
- **If an exact cost is ever added later**, replace the estimated cost C with the real margin (Price – Cost) and the same formula becomes exact — no other change.